

Операционные системы

Командная оболочка Midnight Commander

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Цели и задачи работы

Процесс выполнения лабораторной работы

Выводы по проделанной работе

1. Цели и задачи работы



Ознакомление с файловой системой Linux, её структурой, именами и содержанием каталогов. Приобретение практических навыков по применению команд для работы с файлами и каталогами, по управлению процессами, по проверке использования диска и обслуживанию файловой системы.



- 1 Изучить возможности Midnight Commander
- 2 Изучить редактор Midnight Commander

2. Процесс выполнения лабораторной работы

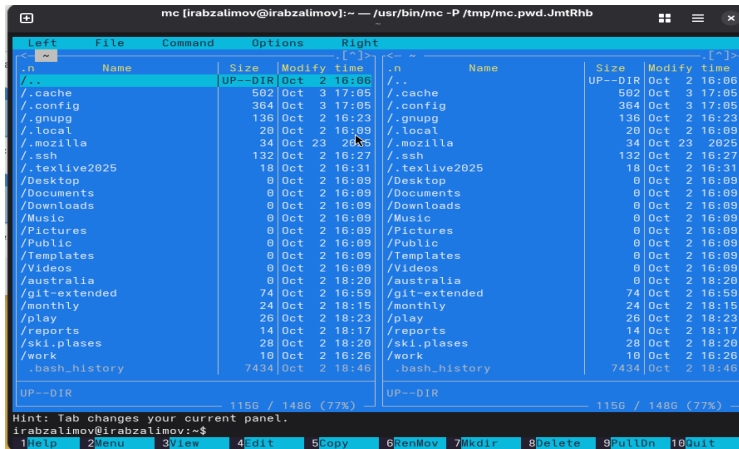


Рисунок 1: Запуск mc

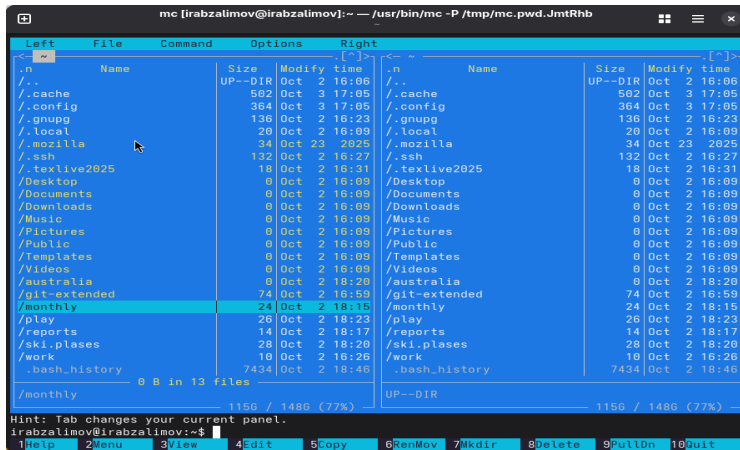


Рисунок 2: Выделение



```
mc [irabzalimov@irabzalimov]: ~ — /usr/bin/mc -P /tmp/mc.pwd.JmtRhb
```

Left	File	Command	Options	Right
< ~				. [^]>
	Name	Size	Modify time	
./	mozilla	34	Oct 23 2025	UP--DIR
./	ssh	132	Oct 2 16:27	502 Oct 3 17:05
./	texlive2025	18	Oct 2 16:31	364 Oct 3 17:05
/	Desktop	0	Oct 2 16:09	136 Oct 2 16:23
/	Documents	0	Oct 2 16:09	20 Oct 2 16:09
/	Downloads	0	Oct 2 16:09	34 Oct 23 2025
/	Music	0	Oct 2 16:09	132 Oct 2 16:27
/	Pictures	0	Oct 2 16:09	18 Oct 2 16:31
/	Public	0	Oct 2 16:09	0 Oct 2 16:09
/	Templates	0	Oct 2 16:09	0 Oct 2 16:09
/	Videos	0	Oct 2 16:09	0 Oct 2 16:09
/	australia	0	Oct 2 18:20	0 Oct 2 16:09
/	git-extended	74	Oct 2 16:59	0 Oct 2 16:09
/	monthly	24	Oct 2 18:15	0 Oct 2 16:09
/	play	26	Oct 2 18:23	0 Oct 2 16:09
/	reports	14	Oct 2 18:17	0 Oct 2 16:09
/	ski.plases	28	Oct 2 18:20	0 Oct 2 18:20
/	work	10	Oct 2 16:26	74 Oct 2 16:59
.	bash_history	7434	Oct 2 18:46	24 Oct 2 18:15
.	bash_logout	18	Jul 23 2025	26 Oct 2 18:23
.	bash_profile	144	Jul 23 2025	14 Oct 2 18:17
.	bashrc	681	Oct 2 16:48	28 Oct 2 18:20
.	emacs	334	Aug 28 03:00	10 Oct 2 16:26
.	gitconfig	236	Oct 2 17:06	7434 Oct 2 18:46
/monthly				UP--DIR
		1156 / 1486 (77%)		1156 / 1486 (77%)

Hint: To change directory halfway through typing a command, use M-c (quick cd).

```
irabzalimov@irabzalimov:~$
```

1 Help 2 Menu 3 View 4 Edit 5 Copy 6 RenMov 7 Mkdir 8 Delete 9 PullDn 10 Quit

Рисунок 3: Отмена

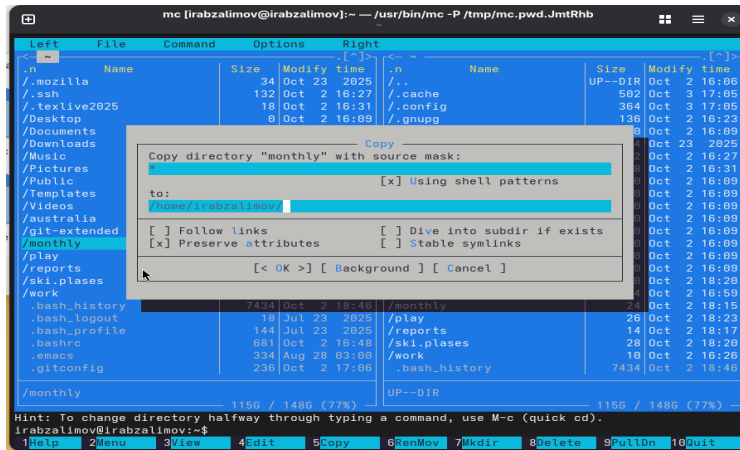


Рисунок 4: Копирование

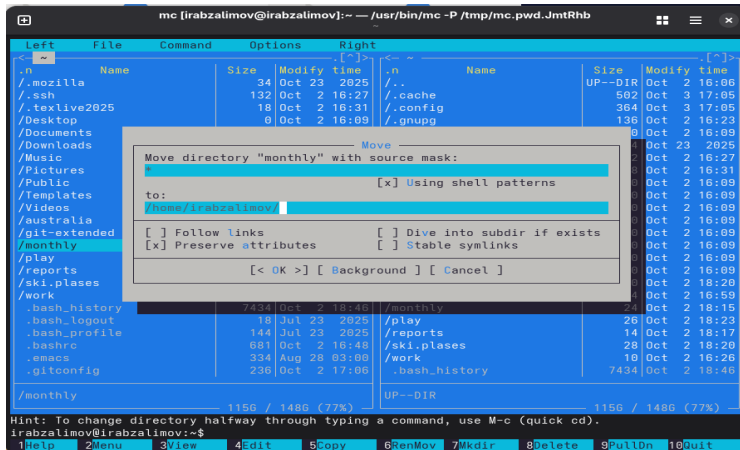


Рисунок 5: Перемещение



```
mc [irabzalimov@irabzalimov]:~ — /usr/bin/mc -P /tmp/mc.pwd.JmtRhb
```

Left	File	Command	Options	Right																																																																																																				
< ~				[^]>																																																																																																				
<table border="1"> <thead> <tr> <th>Size</th> <th>Name</th> <th>Modify</th> <th>time</th> </tr> </thead> <tbody> <tr><td>34</td><td>/.mozilla</td><td>Oct 23</td><td>2025</td></tr> <tr><td>132</td><td>/.ssh</td><td>Oct 2</td><td>16:27</td></tr> <tr><td>18</td><td>/.texlive2025</td><td>Oct 2</td><td>16:31</td></tr> <tr><td>0</td><td>/Desktop</td><td>Oct 2</td><td>16:09</td></tr> <tr><td>0</td><td>/Documents</td><td>Oct 2</td><td>16:09</td></tr> <tr><td>0</td><td>/Downloads</td><td>Oct 2</td><td>16:09</td></tr> <tr><td>0</td><td>/Music</td><td>Oct 2</td><td>16:09</td></tr> <tr><td>0</td><td>/Pictures</td><td>Oct 2</td><td>16:09</td></tr> <tr><td>0</td><td>/Public</td><td>Oct 2</td><td>16:09</td></tr> <tr><td>0</td><td>/Templates</td><td>Oct 2</td><td>16:09</td></tr> <tr><td>0</td><td>/Videos</td><td>Oct 2</td><td>16:09</td></tr> <tr><td>0</td><td>/australia</td><td>Oct 2</td><td>18:20</td></tr> <tr><td>74</td><td>/git-extended</td><td>Oct 2</td><td>16:59</td></tr> <tr><td>24</td><td>/monthly</td><td>Oct 2</td><td>18:15</td></tr> <tr><td>26</td><td>/play</td><td>Oct 2</td><td>18:23</td></tr> <tr><td>14</td><td>/reports</td><td>Oct 2</td><td>18:17</td></tr> <tr><td>28</td><td>/ski.plases</td><td>Oct 2</td><td>18:20</td></tr> <tr><td>10</td><td>/work</td><td>Oct 2</td><td>16:26</td></tr> <tr><td>7434</td><td>.bash_history</td><td>Oct 2</td><td>18:46</td></tr> <tr><td>18</td><td>.bash_logout</td><td>Jul 23</td><td>2025</td></tr> <tr><td>144</td><td>.bash_profile</td><td>Jul 23</td><td>2025</td></tr> <tr><td>681</td><td>.bashrc</td><td>Oct 2</td><td>16:48</td></tr> <tr><td>334</td><td>.emacs</td><td>Aug 28</td><td>03:00</td></tr> <tr><td>236</td><td>.gitconfig</td><td>Oct 2</td><td>17:06</td></tr> </tbody> </table>				Size	Name	Modify	time	34	/.mozilla	Oct 23	2025	132	/.ssh	Oct 2	16:27	18	/.texlive2025	Oct 2	16:31	0	/Desktop	Oct 2	16:09	0	/Documents	Oct 2	16:09	0	/Downloads	Oct 2	16:09	0	/Music	Oct 2	16:09	0	/Pictures	Oct 2	16:09	0	/Public	Oct 2	16:09	0	/Templates	Oct 2	16:09	0	/Videos	Oct 2	16:09	0	/australia	Oct 2	18:20	74	/git-extended	Oct 2	16:59	24	/monthly	Oct 2	18:15	26	/play	Oct 2	18:23	14	/reports	Oct 2	18:17	28	/ski.plases	Oct 2	18:20	10	/work	Oct 2	16:26	7434	.bash_history	Oct 2	18:46	18	.bash_logout	Jul 23	2025	144	.bash_profile	Jul 23	2025	681	.bashrc	Oct 2	16:48	334	.emacs	Aug 28	03:00	236	.gitconfig	Oct 2	17:06	Information Midnight Commander 4.8.33 File: monthly Location: 2Ch:67D5A8h Mode: drwx--x--x (0711) Attributes: ----- Links: 1 Owner: irabzalimov/irabzalimov Size: 24 (0 blocks) Changed: Oct 2 18:18 Modified: Oct 2 18:15 Accessed: Oct 2 18:31 Filesystem: /home Device: /dev/nvme0n1p3 Type: btrfs (22h) Free space: 1156 / 1486 (77%) No node information
Size	Name	Modify	time																																																																																																					
34	/.mozilla	Oct 23	2025																																																																																																					
132	/.ssh	Oct 2	16:27																																																																																																					
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0	/Videos	Oct 2	16:09																																																																																																					
0	/australia	Oct 2	18:20																																																																																																					
74	/git-extended	Oct 2	16:59																																																																																																					
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14	/reports	Oct 2	18:17																																																																																																					
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236	.gitconfig	Oct 2	17:06																																																																																																					
/monthly				1156 / 1486 (77%)																																																																																																				

Hint: To change directory halfway through typing a command, use M-c (quick cd).

```
irabzalimov@irabzalimov:~$
```

1Help 2Menu 3View 4Edit 5Copy 6RenMov 7Mkdir 8Delete 9PullDn 10Quit

Рисунок 6: Информация



The screenshot shows the Midnight Commander (MC) interface. The title bar indicates the user is `irabzalimov@irabzalimov` in the directory `~/usr/bin/mc -P /tmp/mc.pwd.JmtRhb`. The interface is split into two main panes: a left pane for file listing and a right pane for file editing.

Left Pane (File List):

File	Command	Options	Right
.n	Name	Size	Modify time
./mozilla		34	Oct 23 2025
./ssh		132	Oct 2 16:27
./texlive2025		18	Oct 2 16:31
/Desktop		0	Oct 2 16:09
/Documents		0	Oct 2 16:09
/Downloads		0	Oct 2 16:09
/Music		0	Oct 2 16:09
/Pictures		0	Oct 2 16:09
/Public		0	Oct 2 16:09
/Templates		0	Oct 2 16:09
/Videos		0	Oct 2 16:09
/australia		0	Oct 2 18:20
/git-extended		74	Oct 2 16:59
/monthly		24	Oct 2 18:15
/play		26	Oct 2 18:23
/reports		14	Oct 2 18:17
/ski.plases		28	Oct 2 18:20
/work		10	Oct 2 16:26
.bash_history		7434	Oct 2 18:46
.bash_logout		18	Jul 23 2025
.bash_profile		144	Jul 23 2025
.bashrc		681	Oct 2 16:48
.emacs		334	Aug 28 03:00
.gitconfig		236	Oct 2 17:06
.bash_profile			

At the bottom of the left pane, it shows `1156 / 1486 (77%)`.

Right Pane (File Editor):

The right pane shows the contents of the selected file, `.bash_profile`. The editor displays the following text:

```
/home/~rofile 144/144 100%
# .bash_profile

# Get the aliases and functions
if [ -f ~/.bashrc ]; then
    . ~/.bashrc
fi

# User specific environment and startup program
s
```

Below the editor, a hint is displayed: `Hint: To change directory halfway through typing a command, use M-c (quick cd).`

The status bar at the bottom shows the user `irabzalimov@irabzalimov:~$` and a list of keyboard shortcuts: `1Help 2Menu 3View 4Edit 5Copy 6RenMov 7Mkdir 8Delete 9PullDn 10Quit`.

Рисунок 7: Быстрый просмотр

mc [irabzalimov@irabzalimov]: ~ — /usr/bin/mc -P /tmp/mc.pwd.JmtRhb

Left	File	Command	Options	Right
< ^>				[^]>
.n	Name	Size	Modify time	Information
/.mozilla		34	Oct 23 2025	Midnight Commander 4.8.33
/.ssh		132	Oct 2 16:27	File: .bash_profile
/.texlive2025		18	Oct 2 16:31	Location: 2Ch:672110h
/Desktop		0	Oct 2 16:09	Mode: -rw-r--r-- (8644)
/Documents		0	Oct 2 16:09	Attributes: -----
/Downloads		0	Oct 2 16:09	Links: 1
/Music		0	Oct 2 16:09	Owner: irabzalimov/irabzalimov
/Pictures		0	Oct 2 16:09	Size: 144 (8 blocks)
/Public		0	Oct 2 16:09	Changed: Oct 2 16:06
/Templates		0	Oct 2 16:09	Modified: Jul 23 2025
/Videos		0	Oct 2 16:09	Accessed: Oct 3 17:02
/australia		0	Oct 2 18:20	Filesystem: /home
/git-extended		74	Oct 2 16:59	Device: /dev/nvme0n1p3
/monthly		24	Oct 2 18:15	Type: btrfs (22h)
/play		26	Oct 2 18:23	Free space: 115G / 148G (77%)
/reports		14	Oct 2 18:17	No node information
/ski.plases		28	Oct 2 18:20	
/work		10	Oct 2 16:26	
.bash_history		7434	Oct 2 18:46	
.bash_logout		18	Jul 23 2025	
.bash_profile		144	Jul 23 2025	
.bashrc		681	Oct 2 16:48	
.emacs		334	Aug 28 03:00	
.gitconfig		236	Oct 2 17:06	
.bash_profile				

115G / 148G (77%)

Hint: To change directory halfway through typing a command, use M-c (quick cd).

irabzalimov@irabzalimov:~\$

1Help 2Menu 3View 4Edit 5Copy 6RenMov 7Mkdir 8Delete 9PullDn 10Quit

Рисунок 8: Информация





```
mc [irabzalimov@irabzalimov]:~/work/study/2025-2026/Операционные системы/2026-1--st...
~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro

/home/irabzalimov/work/study/2025-2026-1--study--os-intro/Makefile 1807/1821 98%
SHELL := /bin/bash
COURSE =

.PHONY: all clean

all: help

help:
    @echo 'Usage:'
    @echo '  make <target>'
    @echo
    @echo 'Targets:'
    @grep -E '^[a-zA-Z_0-9-]+:.*?###.*$$' $(MAKEFILE_LIST) | grep -v '###' | sort | cut -d: -f1-
1- | awk 'BEGIN {FS = ".*?###"}; {printf " \033[36m%-30s\033[0m %s\n", $$1, $$2}'
    @grep -E '###.*' $(MAKEFILE_LIST) | cut -d: -f2- | awk 'BEGIN {FS = "###"}; {printf "%s\n", $$1, $$2}'
    @grep -E '^[a-zA-Z_0-9-]+:.*?###.*$$' $(MAKEFILE_LIST) | sort | cut -d: -f2- | awk 'BEGIN
{FS = ".*?###"}; {printf " \033[36m%-30s\033[0m %s\n", $$1, $$2}'
    @echo

list: ## List of courses
    @. ./template/config/script/list-courses

prepare: ## Generate directories structure
    @. ./template/config/script/prepare
    @touch prepare

submodule: ## Update submodules
    git submodule update --init --recursive
    git submodule foreach 'git fetch origin; git checkout $(git rev-parse --abbrev-ref HEAD);
git reset --hard origin/$(git rev-parse --abbrev-ref HEAD); git submodule update --recursive; gi
1Help 2UnWrap 3Quit 4Hex 5Goto 6 7Search 8Raw 9Format 10Quit
```

Рисунок 10: Просмотр содержимого текстового файла

```
mc [irabzalimov@irabzalimov]:~/work/study/2025-2026/Операционные системы/2026-1--st...
~/work/study/2025-2026/Операционные системы/2026-1--study--os-Intro

Makefile  [----]  0 L: [ 1+ 0  1/ 28]  *(0  /1021b) 0083 0x053  [*][X]

SHELL := /bin/bash
COURSE =.

.PHONY: all clean

all: help

help:
<----->@echo 'Usage:'
<----->@echo '  make <target>'
<----->@echo.
<----->@echo 'Targets:'
<----->@grep -E '^([a-zA-Z_0-9-]+).*?###.*$$' $(MAKEFILE_LIST) | grep -v '###' | sort | cut -d: -f
<----->@grep -E '^###.*' $(MAKEFILE_LIST) | cut -d' ' -f2- | awk 'BEGIN {FS = "###"}; {printf "%s
<----->@grep -E '^([a-zA-Z_0-9-]+).*?###.*$$' $(MAKEFILE_LIST) | sort | cut -d: -f2- | awk 'BEGIN
<----->@echo

list:<-->## List of courses
<----->@. ./template/config/script/list-courses

prepare:<----->## Generate directories structure
<----->@. ./template/config/script/prepare
<----->@touch prepare

submodule:<----->## Update submules
<----->git submodule update --init --recursive
<----->git submodule foreach 'git fetch origin; git checkout $$ (git rev-parse --abbrev-ref HEAD);

1Help  2Save  3Mark  4Replac  5Copy  6Move  7Search  8Delete  9PullDn  10Quit
```

Рисунок 11: Отредактируем содержимое текстового файла без сохранения результатов

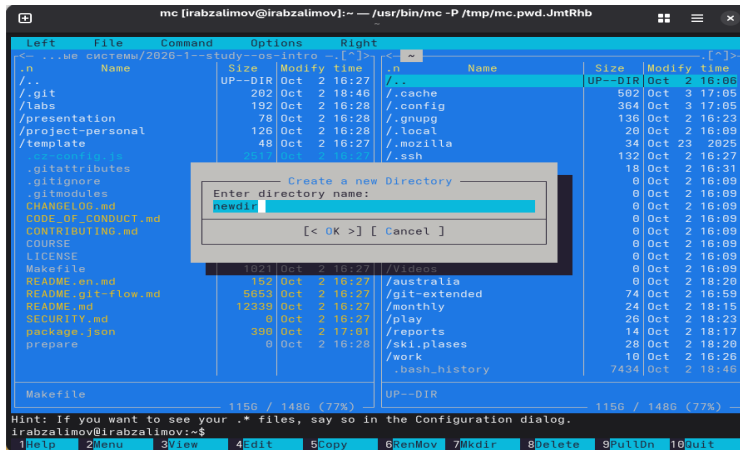


Рисунок 12: Создание каталога

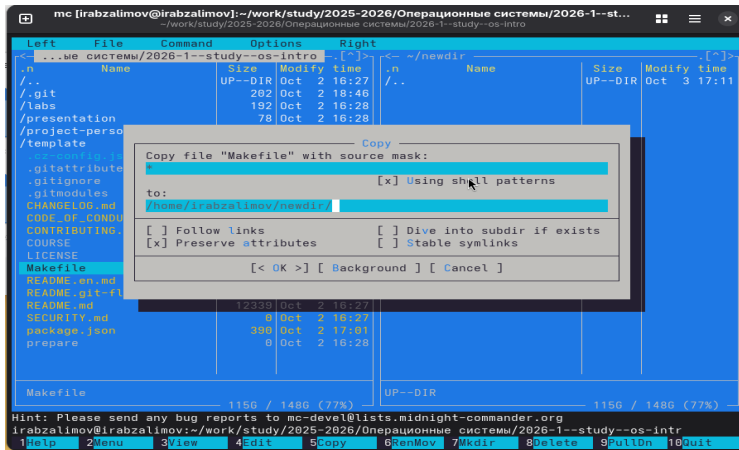


Рисунок 13: Копирование в файлов в созданный каталог







```
mc [irabzalimov@irabzalimov]:~ — /usr/bin/mc -P /tmp/mc.pwd.JmtRh6
```

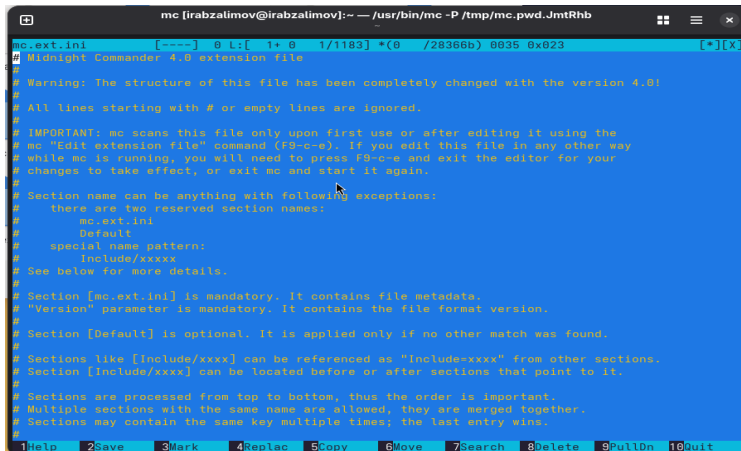
Left	File	Command	Options	Right
~				~/newdir
..	Name	Size	Modify time	..
UP--DIR				UP--DIR
./cache		502	Oct 3 17:05	Makefile
./config		364	Oct 3 17:05	
./gnupg		136	Oct 2 16:23	
./local		20	Oct 2 16:09	
./mozilla		34	Oct 23 2025	
./ssh		132	Oct 2 16:27	
./texlive2025		18	Oct 2 16:31	
/Desktop		0	Oct 2 16:09	
/Documents		0	Oct 2 16:09	
/Downloads		0	Oct 2 16:09	
/Music		0	Oct 2 16:09	
/Pictures		0	Oct 2 16:09	
/Public		0	Oct 2 16:09	
/Templates		0	Oct 2 16:09	
/Videos		0	Oct 2 16:09	
/australia		0	Oct 2 18:20	
/git-extended		74	Oct 2 16:59	
/monthly		24	Oct 2 18:15	
/newdir		16	Oct 3 17:11	
/play		26	Oct 2 18:23	
/reports		14	Oct 2 18:17	
/ski.plases		28	Oct 2 18:20	
/work		10	Oct 2 16:26	
UP--DIR				UP--DIR
115G / 148G (77%)				115G / 148G (77%)

Hint: Please send any bug reports to mc-devel@lists.midnight-commander.org

```
irabzalimov@irabzalimov:~$ cd
```

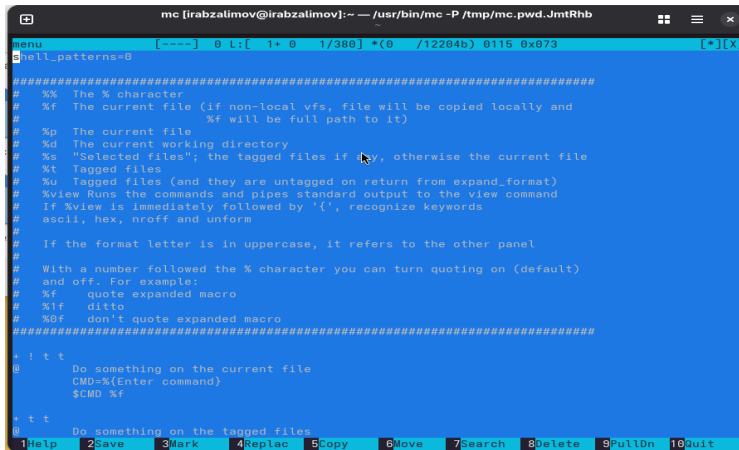
1Help 2Menu 3View 4Edit 5Copy 6RenMov 7Mkdir 8Delete 9PullDn 10Quit

Рисунок 16: Переход в домашний каталог



```
mc [irabzalimov@irabzalimov]:~ — /usr/bin/mc -P /tmp/mc.pwd.JmtRhb
mc.ext.ini [----] 0 L:[ 1+ 0 1/1183] *(0 /28366b) 0035 0x023 [*](x)
# Midnight Commander 4.8 extension file
#
# Warning: The structure of this file has been completely changed with the version 4.8!
#
# All lines starting with # or empty lines are ignored.
#
# IMPORTANT: mc scans this file only upon first use or after editing it using the
# mc "Edit extension file" command (F9-c-e). If you edit this file in any other way
# while mc is running, you will need to press F9-c-e and exit the editor for your
# changes to take effect, or exit mc and start it again.
#
# Section name can be anything with following exceptions:
#   there are two reserved section names:
#     mc.ext.ini
#     Default
#   special name pattern:
#     Include/xxxxx
# See below for more details.
#
# Section [mc.ext.ini] is mandatory. It contains file metadata.
# "Version" parameter is mandatory. It contains the file format version.
#
# Section [Default] is optional. It is applied only if no other match was found.
#
# Sections like [Include/xxxx] can be referenced as "Include=xxxx" from other sections.
# Section [Include/xxxx] can be located before or after sections that point to it.
#
# Sections are processed from top to bottom, thus the order is important.
# Multiple sections with the same name are allowed, they are merged together.
# Sections may contain the same key multiple times; the last entry wins.
#
1Help 2Save 3Mark 4Replac 5Copy 6Move 7Search 8Delete 9PullDn 10Quit
```

Рисунок 17: Просмотр файла расширений



The screenshot shows the Midnight Commander (MC) application window. The title bar reads "mc [irabzalimov@irabzalimov]: ~ — /usr/bin/mc -P /tmp/mc.pwd.JmtRhb". The main window has a blue background with white text. At the top, a status bar shows "menu [----] 0 L:[1+ 0 1/380] *(0 /12204b) 0115 0x073 [*][X]". Below this, the text "hell_patterns=0" is visible. The main content area displays a list of menu items, each preceded by a hash symbol (#). The items are: "% The % character", "%f The current file (if non-local vfs, file will be copied locally and %f will be full path to it)", "%p The current file", "%d The current working directory", "%s \"Selected files\"; the tagged files if %y, otherwise the current file", "%t Tagged files", "%u Tagged files (and they are untagged on return from expand_format)", "%view Runs the commands and pipes standard output to the view command", "If %view is immediately followed by '{', recognize keywords", "ascii, hex, nroff and unform", "If the format letter is in uppercase, it refers to the other panel", "With a number followed the % character you can turn quoting on (default) and off. For example:", "%f quote expanded macro", "%lf ditto", "%0f don't quote expanded macro". Below these items, there are two sections: "+ ! t t" and "+ t t", each followed by a description and a command prompt. The bottom of the window features a menu bar with 10 items: 1Help, 2Save, 3Mark, 4Replac, 5Copy, 6Move, 7Search, 8Delete, 9PullDn, 10Quit.

```
mc [irabzalimov@irabzalimov]: ~ — /usr/bin/mc -P /tmp/mc.pwd.JmtRhb
menu [----] 0 L:[ 1+ 0 1/380] *(0 /12204b) 0115 0x073 [*][X]
hell_patterns=0

#####
# % The % character
# %f The current file (if non-local vfs, file will be copied locally and
# %f will be full path to it)
# %p The current file
# %d The current working directory
# %s "Selected files"; the tagged files if %y, otherwise the current file
# %t Tagged files
# %u Tagged files (and they are untagged on return from expand_format)
# %view Runs the commands and pipes standard output to the view command
# If %view is immediately followed by '{', recognize keywords
# ascii, hex, nroff and unform
#
# If the format letter is in uppercase, it refers to the other panel
#
# With a number followed the % character you can turn quoting on (default)
# and off. For example:
# %f quote expanded macro
# %lf ditto
# %0f don't quote expanded macro
#####

+ ! t t
@ Do something on the current file
  CMD=%{Enter command}
  $CMD %f

+ t t
@ Do something on the tagged files

1Help 2Save 3Mark 4Replac 5Copy 6Move 7Search 8Delete 9PullDn 10Quit
```

Рисунок 18: Просмотр файла меню

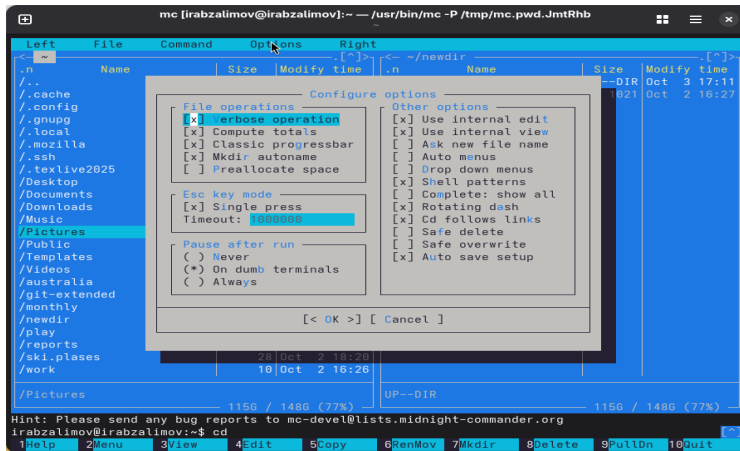


Рисунок 19: Конфигурация

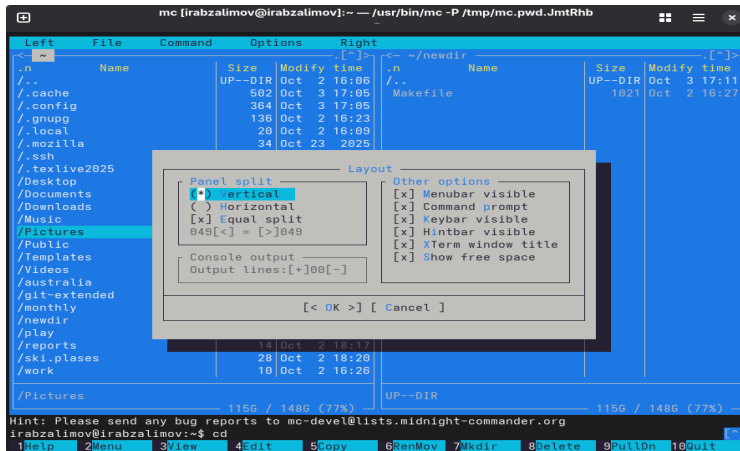


Рисунок 20: Внешний вид

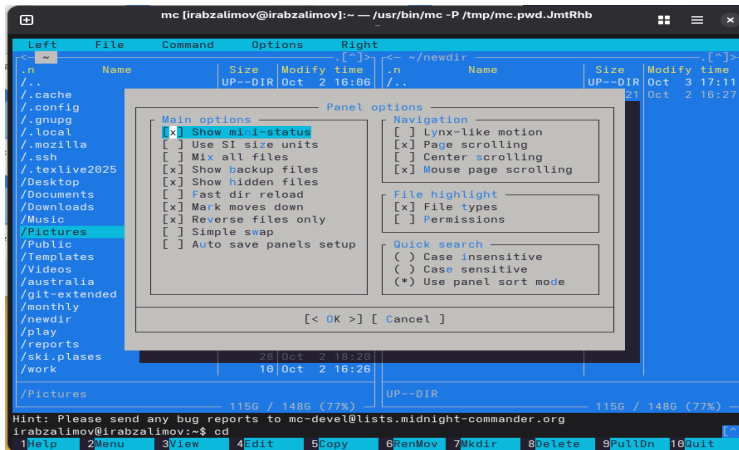


Рисунок 21: Настройки панелей

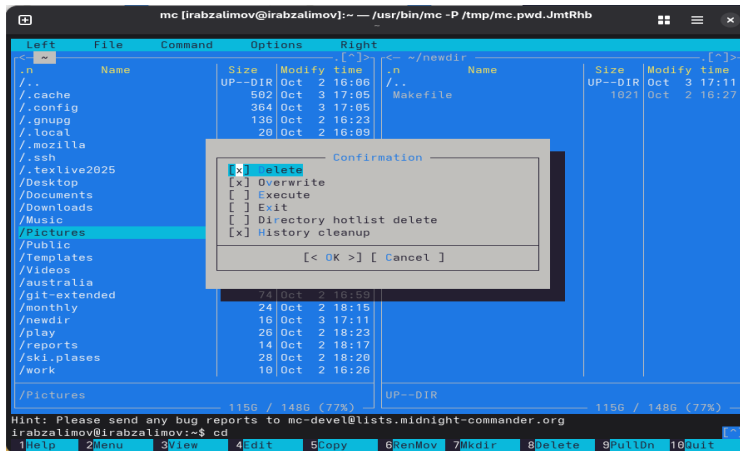


Рисунок 22: Подтверждение

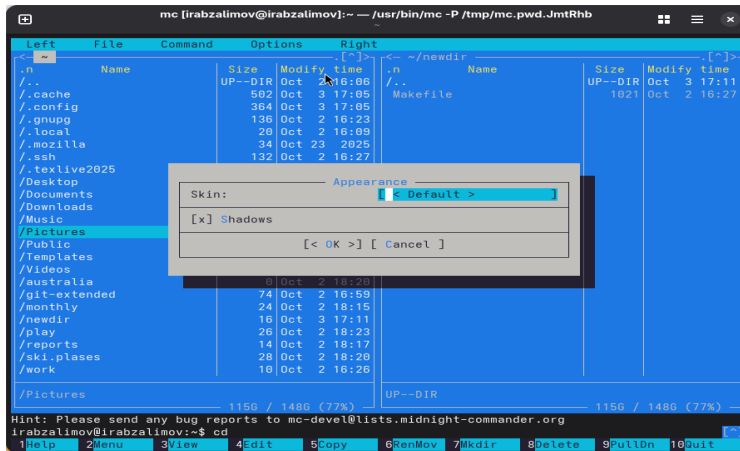


Рисунок 23: Оформление

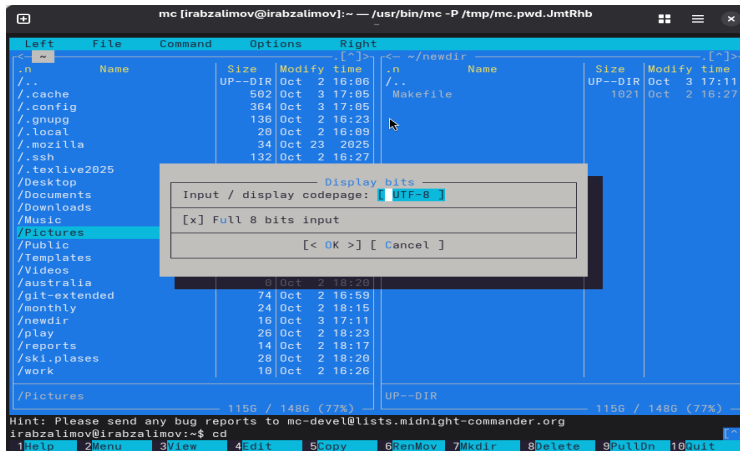


Рисунок 24: Кодировка символов

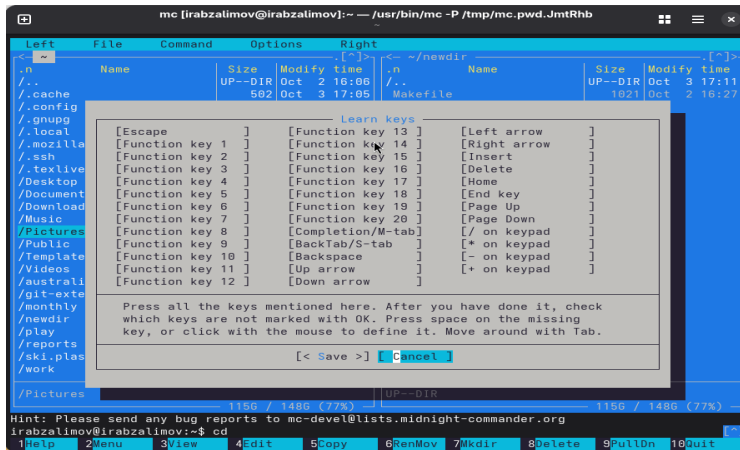


Рисунок 25: Распознавание клавиш



```
mc [irabzalimov@irabzalimov]:~/newdir — /usr/bin/mc -P /tmp/mc.pwd.JmtRhb
~/newdir
Makefile  [----] 13 L:[ 1+15 16/ 28] *(580 /1021b) 0010 0x00A [*][X]
SHELL := /bin/bash
COURSE =.

.PHONY: all clean

all: help

help:
<----->@echo 'Usage:'
<----->@echo '  make <target>'
<----->@echo.
<----->@echo 'Targets:'
<----->@grep -E '^[a-zA-Z_0-9-]+:.*?##.*$$' $(MAKEFILE_LIST) | grep -v '###' | sort | cut -d: -f1
<----->@grep -E '^[a-zA-Z_0-9-]+:.*?##.*$$' $(MAKEFILE_LIST) | cut -d: -f2- | awk 'BEGIN {FS = "###"}; {printf "%s\n", $2}'
<----->@grep -E '^[a-zA-Z_0-9-]+:.*?##.*$$' $(MAKEFILE_LIST) | sort | cut -d: -f2- | awk 'BEGIN {FS = "###"}; {printf "%s\n", $2}'
<----->@echo

list:<-->## List of courses
<----->@. ./template/config/script/list-courses

prepare:<----->## Generate directories structure
<----->@. ./template/config/script/prepare
<----->@touch prepare

submodule:<----->## Update submodules
<----->git submodule update --init --recursive
<----->git submodule foreach 'git fetch origin; git checkout $$ (git rev-parse --abbrev-ref HEAD);'
```

Рисунок 26: Файл с текстом



```
mc [irabzalimov@irabzalimov]:~/newdir — /usr/bin/mc -P /tmp/mc.pwd.JmtRhb
~ /newdir

Makefile  [-M--]  0 L:[ 1+15 16/ 24] *(574 / 946b) 0010 0x00A  [*][X]
SHELL := /bin/bash
COURSE =.

.PHONY: all clean

all: help

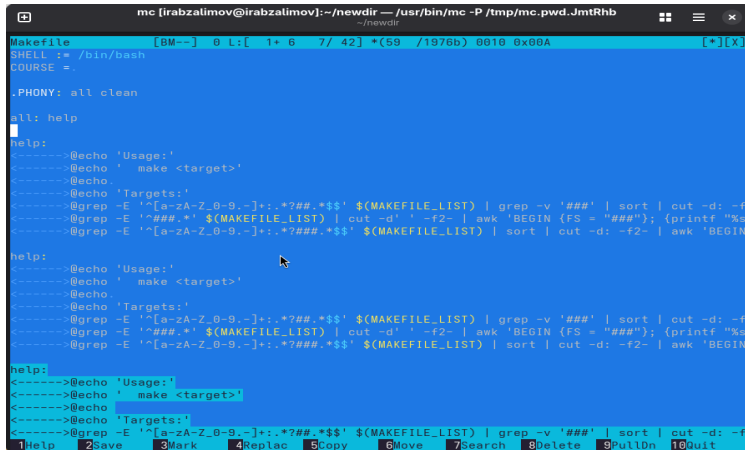
help:
<----->@echo 'Usage:'
<----->@echo '  make <target>'
<----->@echo.
<----->@echo 'Targets:'
<----->@grep -E '^[a-zA-Z_0-9-]+:.*?###.*$$' $(MAKEFILE_LIST) | grep -v '###' | sort | cut -d: -f
<----->@grep -E '^[^###.*] $(MAKEFILE_LIST) | cut -d' ' -f2- | awk 'BEGIN (FS = "###"); {printf "%s
<----->@grep -E '^[a-zA-Z_0-9-]+:.*?###.*$$' $(MAKEFILE_LIST) | sort | cut -d: -f2- | awk 'BEGIN

prepare:<----->## Generate directory structure
<----->@. ./template/config/script/prepare
<----->@touch prepare

submodule:<----->## Update submules
<----->git submodule update --init --recursive
<----->git submodule foreach 'git fetch origin; git checkout $$ (git rev-parse --abbrev-ref HEAD);

1Help  2Save  3Mark  4Replac  5Copy  6Move  7Search  8Delete  9PullDn  10Quit
```

Рисунок 27: Файл с текстом



```
mc [irabzalimov@irabzalimov]:~/newdir — /usr/bin/mc -P /tmp/mc.pwd.JmtRhb
~/newdir
Makefile  [BM--]  0 L: [ 1+ 6  7/ 42] +(59 /1976b) 0010 0x00A  [+][X]
SHELL := /bin/bash
COURSE =.

.PHONY: all clean

all: help

help:
<----->@echo 'Usage:'
<----->@echo '  make <target>'
<----->@echo.
<----->@echo 'Targets:'
<----->@grep -E '^[a-zA-Z_0-9-]+:.*?##.*$$' $(MAKEFILE_LIST) | grep -v '###' | sort | cut -d: -f
<----->@grep -E '^###.*' $(MAKEFILE_LIST) | cut -d' ' -f2- | awk 'BEGIN {FS = "###"}; {printf "%s
<----->@grep -E '^[a-zA-Z_0-9-]+:.*?##.*$$' $(MAKEFILE_LIST) | sort | cut -d: -f2- | awk 'BEGIN

help:
<----->@echo 'Usage:'
<----->@echo '  make <target>'
<----->@echo.
<----->@echo 'Targets:'
<----->@grep -E '^[a-zA-Z_0-9-]+:.*?##.*$$' $(MAKEFILE_LIST) | grep -v '###' | sort | cut -d: -f
<----->@grep -E '^###.*' $(MAKEFILE_LIST) | cut -d' ' -f2- | awk 'BEGIN {FS = "###"}; {printf "%s
<----->@grep -E '^[a-zA-Z_0-9-]+:.*?##.*$$' $(MAKEFILE_LIST) | sort | cut -d: -f2- | awk 'BEGIN

help:
<----->@echo 'Usage:'
<----->@echo '  make <target>'
<----->@echo.
<----->@echo 'Targets:'
<----->@grep -E '^[a-zA-Z_0-9-]+:.*?##.*$$' $(MAKEFILE_LIST) | grep -v '###' | sort | cut -d: -f
1Help  2Save  3Mark  4Replac  5Copy  6Move  7Search  8Delete  9PullDn 10Quit
```

Рисунок 28: Копирование фрагмента

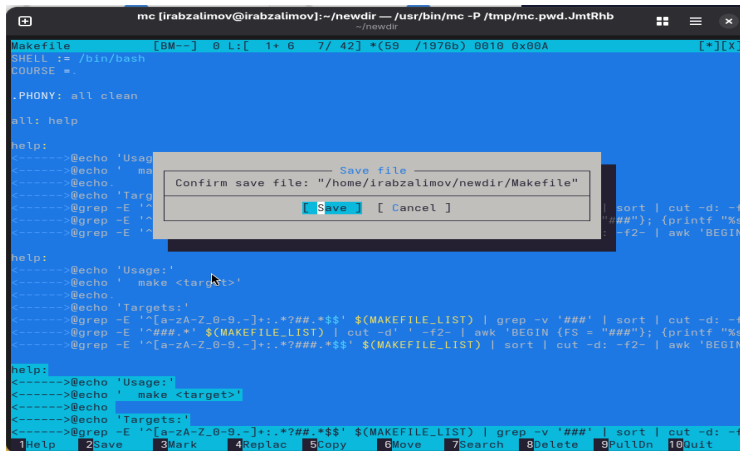


Рисунок 29: Сохранение



```
mc [irabzalimov@irabzalimov]:~/newdir — /usr/bin/mc -P /tmp/mc.pwd.JmtRhb
~ /newdir

Makefile  [BM--]  0 L:[ 1+ 6 7/ 24] *(59 / 946b) 0010 0x00A  [*][X]
SHELL := /bin/bash
COURSE =.

.PHONY: all clean

all: help

help:
<----->@echo 'Usage:'
<----->@echo ' make <target>'
<----->@echo
<----->@echo 'Targets:'
<----->@grep -E '^[a-zA-Z_0-9.]+:.*?###.*$$' $(MAKEFILE_LIST) | grep -v '###' | sort | cut -d: -f2-
<----->@grep -E '^###.*' $(MAKEFILE_LIST) | cut -d' ' -f2- | awk 'BEGIN {FS = "###"}; {printf "%s\n", $0}'
<----->@grep -E '^[a-zA-Z_0-9.]+:.*?###.*$$' $(MAKEFILE_LIST) | sort | cut -d: -f2- | awk 'BEGIN

prepare:<----->## Generate directories structure
<----->@. ./template/config/script/prepare
<----->@touch prepare

submodule:<----->## Update submules
<----->git submodule update --init --recursive
<----->git submodule foreach 'git fetch origin; git checkout $$ (git rev-parse --abbrev-ref HEAD);

1Help  2Save  3Mark  4Replac  5Copy  6Move  7Search  8Delete  9PullDn  10Quit
```

Рисунок 30: Отмена

```
mc [irabzalimov@irabzalimov]:~/newdir — /usr/bin/mc -P /tmp/mc.pwd.JmtRhb
~[newdir]
Makefile  [-M--]  0 L:[ 1+23 24/ 24] *(946 / 946b) <E0F>  [*][X]
SHELL := /bin/bash
COURSE = .
.PHONY: all clean
all: help
help:
<----->@echo 'Usage:'
<----->@echo '  make <target>'
<----->@echo .
<----->@echo 'Targets:'
<----->@grep -E '^([a-zA-Z_0-9.-]+:.*?###.*$$)' $(MAKEFILE_LIST) | grep -v '###' | sort | cut -d: -f1
<----->@grep -E '^###.*' $(MAKEFILE_LIST) | cut -d' ' -f2- | awk 'BEGIN {FS = "###"}; {printf "%s\n", $0}'
<----->@grep -E '^([a-zA-Z_0-9.-]+:.*?###.*$$)' $(MAKEFILE_LIST) | sort | cut -d: -f2- | awk 'BEGIN
prepare:<----->## Generate directories structure
<----->@. ./template/config/script/prepare
<----->@touch prepare
submodule:<----->## Update submules
<----->git submodule update --init --recursive
<----->git submodule foreach 'git fetch origin; git checkout $$ (git rev-parse --abbrev-ref HEAD);
1Help  2Save  3Mark  4Replac  5Copy  6Move  7Search  8Delete  9PullDn  10Quit
```

Рисунок 31: Переход в конец файла



```
mc [irabzalimov@irabzalimov]:~/newdir — /usr/bin/mc -P /tmp/mc.pwd.JmtRhb
~/newdir
Makefile      [-M-0]  0 L:[ 1+ 0  1/ 24] *(0  / 946b) 0083 0x053  [*][X]
SHELL := /bin/bash
COURSE =
.PHONY: all clean
all: help
help:
<----->@echo 'Usage:'
<----->@echo '  make <target>'
<----->@echo
<----->@echo 'Targets:'
<----->@grep -E '^([a-zA-Z_0-9-]+).*?##.*$$' $(MAKEFILE_LIST) | grep -v '###' | sort | cut -d: -f1
<----->@grep -E '^###.*' $(MAKEFILE_LIST) | cut -d' ' -f2- | awk 'BEGIN {FS = "###"}; {printf "%s\n", $2}'
<----->@grep -E '^([a-zA-Z_0-9-]+).*?##.*$$' $(MAKEFILE_LIST) | sort | cut -d: -f2- | awk 'BEGIN
prepare:<----->## Generate directories structure
<----->@. ./template/config/script/prepare
<----->@touch prepare
submodule:<----->## Update submules
<----->git submodule update --init --recursive
<----->git submodule foreach 'git fetch origin; git checkout $$ (git rev-parse --abbrev-ref HEAD);
1Help  2Save  3Mark  4Replac  5Copy  6Move  7Search  8Delete  9PullDn 10Quit
```

Рисунок 32: Переход в начало файла



```
mc [irabzalimov@irabzalimov]: ~/newdir — /usr/bin/mc -P /tmp/mc.pwd.JmtRh6
~ /newdir

klp-diff.c      [----]  0 L:[ 1+ 0 1/2263] *(0 /58157b) 8047 0x82F  [*][X]
// SPDX-License-Identifier: GPL-2.0-or-later
#define _GNU_SOURCE /* memmem() */
#include <unistd.h>
#include <stdlib.h>
#include <string.h>
#include <libgen.h>
#include <stdio.h>
#include <ctype.h>

#include <objtool/objtool.h>
#include <objtool/warn.h>
#include <objtool/arch.h>
#include <objtool/klp.h>
#include <objtool/util.h>
#include <objtool/special.h>

#include <linux/align.h>
#include <linux/objtool_types.h>
#include <linux/livepatch_external.h>
#include <linux/stringify.h>
#include <linux/string.h>
#include <linux/jhash.h>

#define offsetof(TYPE, MEMBER) sizeof((((TYPE *)0)->MEMBER))

struct elfs {
    ----->struct elf *orig, *patched, *out;
    ----->const char *modname;
};

struct export {
1Help  2Save  3Mark  4Replac  5Copy  6Move  7Search  8Delete  9PullDn 10Quit
```

Рисунок 33: Файл с программой

```
mc [irabzalimov@irabzalimov]:~/newdir — /usr/bin/mc -P /tmp/mc.pwd.JmtRh6
~/newdir
klp-diff.c  [----] 13 L: 1+25 26/2263] *(631 /58157b) 0010 0x00A [*][X]
// SPDX-License-Identifier: GPL-2.0-or-later
#define _GNU_SOURCE /* memmem() */
#include <subcmd/parse-options.h>
#include <stdlib.h>
#include <string.h>
#include <libgen.h>
#include <stdio.h>
#include <ctype.h>

#include <objtool/objtool.h>
#include <objtool/warn.h>
#include <objtool/arch.h>
#include <objtool/klp.h>
#include <objtool/util.h>
#include <objtool/special.h>

#include <linux/align.h>
#include <linux/objtool_type.h>
#include <linux/livepatch.h>
#include <linux/stringify.h>
#include <linux/string.h>
#include <linux/jhash.h>

#define sizeof_field(TYPE, MEMBER) sizeof((((TYPE *)0)->MEMBER))

struct elfs {
    <-----> struct elf *orig, *patched, *out;
    <-----> const char *modname;
};

struct export {
1Help 2Save 3Mark 4Replac 5Copy 6Move 7Search 8Delete 9PullDn 10Quit
```

Рисунок 34: Цветовыделение синтаксиса

3. Выводы по проделанной работе



В данной работе мы ознакомились с инструментами поиска файлов и фильтрации текстовых данных. А также приобрели практические навыки по управлению процессами.